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FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND INFORMATICS

DEMAND PREDICTION SYSTEM FOR E_GOVERNMENT SERVICES

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**A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE
IN COMPUTER SCIENCE OF THE UNIVERSITY OF NAIROBI.**

MARCH 2024

DECLARATION

I hereby declare that this project report is entirely my own original work and it has not been submitted for assessment at this or any other university. Materials of work done by other researchers are mentioned by clear references.

Signature.....

Date 14/6/2024

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The project report has been submitted in partial fulfilment of the requirements of the degree of Bachelor of Science in Computer Science at the University of Nairobi with my approval as the University supervisor for this project.

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ABSTRACT

The integration of IT automation by government entities has redefined the citizen experience, driven by the escalating demand for enhanced convenience and efficiency in our digitally-driven society. As we embrace seamless online transactions and instant access to services such as medical appointments and transportation bookings, citizens increasingly expect swift and efficient interactions with their government. However, it is imperative to acknowledge and overcome the challenges associated with the adoption of intelligent automation in the public sector.

This project endeavors to forecast the demand for e-government services facilitated through the Huduma Centers. It begins by succinctly outlining the problem at hand, elucidating the complexities and intricacies involved in anticipating citizen demand for government services. Through a comprehensive review of existing literature, the project examines how similar challenges have been addressed in the past, identifying gaps and areas for improvement.

Employing an Agile methodology, this project adopts an iterative approach to problem-solving, emphasizing adaptability and collaboration throughout the development process. Leveraging a predictive analytics model, the system endeavors to accurately forecast the demand for e-government services, enabling proactive resource allocation and service optimization. Through this innovative approach, the project seeks to enhance the efficiency and responsiveness of government service delivery, ultimately improving citizen satisfaction and fostering a more streamlined and effective public sector ecosystem.

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CHAPTER ONE: INTRODUCTION

1.1 BACKGROUND

Globally, improving public service delivery has received a lot of attention with various reforms being introduced to address problems faced by citizens. The government of Kenya has adopted the process of reengineering the public service sector to enhance performance and efficiency in service delivery (Korir, Rotich , & Bengat, 2015). Huduma Kenya was introduced by the government of Kenya as a strategy for reforming public service sectors, with the objective of ensuring citizens easy access to various government services. However, since its inception, the program continues to face service delivery challenges mainly due to the ineffective use of implemented systems, and management processes used to monitor the organizations' performance. According to Nyongesa, Ondoro, and Ntonga, (2020), it has been noted that Huduma Kenya serves about 30,000 customers against a target of 80,000 per day, and collects Ksh.12 billion against a target of Ksh. 40 billion annually and at the same time, massive data duplications are still experienced. These are some of the challenges that are hampering the performance of E_Government services systems in terms of providing quality and efficient service to the citizens.

1.2 PROBLEM STATEMENT

Public service departments are there to provide services needed by the citizens, as every citizen has the right to quality service (Venables, 2015). According to (Wahida, Ismail J, 2016) Unlike in the private sector where customer satisfaction is the key driver in all the activities of an entity, it is exactly the opposite in the public sector. Public service delivery points in Kenya have witnessed long queues which often leave customers unsatisfied and disappointed. There is evident need for the introduction of the predictive model. There is no ICT support tool with a predictive model to anticipate the demand for services at Huduma Centers. A typical case study; the KCPE 2023 results retrieval incident, as reported by Dennis Musau and CS Machogu (2023), serves as a poignant illustration of a critical gap in the current landscape of e-government services. The case study underscores the absence of a robust tool capable of anticipating and managing the overwhelming demand for government services. The failure to predict and cater to the surge in user demand led to a system breakdown, leaving candidates unable to access their results. This real-world scenario vividly highlights the imperative need for predictive tools to ensure the seamless delivery of e-government services, preventing such operational challenges and ensuring a responsive and reliable service infrastructure.

1.3 GOAL

The goal is to address the existing challenges related to delays in e_government services delivery at Huduma centers.long queues, customer dissatisfaction, and operational breakdowns by anticipating and managing the demand for government services. By leveraging predictive analytics, the project seeks to enhance the efficiency and responsiveness of public service delivery, ensuring a seamless and reliable experience for citizens accessing e-government services.

1.2 OBJECTIVES

1.2.1 System Project Objectives

1. Determine the system requirements.
2. Design the system with predictive model.
3. Implement and test the system.
4. Evaluate the system

Table 1: How the Objectives will be delivered

Objectives	How objectives will be achieved	Deliverables
Determine the system requirements	Acquire data from the Huduma Center database through official channels.	Dataset capturing citizens' appointment booking at Huduma Centers.
Design the system with predictive model	Determine relevant features to be used as inputs and perform feature engineering and feature selection to create other features if needed	Structured data
Implement and test the system.	-Split data to train and test then fit it to multi-linear regression model. -Build the User Interface -Test the model using test data	Predicted values representing the demand for different services at Huduma center.
Evaluate the system	Calculate the model accuracy	R-squared, Mean Squared Error, Root Mean Squared Error values.

1.2.2 Research Objectives

1. Investigate the application of Machine Learning in demand prediction.
2. Investigate the machine learning algorithms that are most effective for demand prediction.

1.2.3 Significance of the study

To improve the experience of customers by reducing waiting times and enhancing the quality-of-service delivery. By anticipating the demand for services, the predictive model will enable Huduma Centers to allocate resources and employee capacity where they are needed in advance

of a crisis arising. This will allow them to respond to citizens faster and before they reach the point of crisis. The predictive model will also enable Huduma Centers to make data-driven decisions that will improve the efficiency of service delivery.

1.2.4 Limitations of the study

The accuracy of the model will heavily rely on the quality and availability of historical data. In some cases, obtaining comprehensive and reliable historical data on service demand may be challenging.

The predictive model assumes a level of consistency and some factors are assumed to be constant in user behavior. However, shifts in user behavior or changes in external factors like extra charges for services, more counties added by the government can impact the prediction of the model.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The goal of the public service by government worldwide is to lead the modernization of service delivery with the aim of providing efficient and reliable services to meet the demands of citizens. E-Government services mature globally with technology, The scope for delivering these services is on the rise leading to attracting entrepreneurs to set up E-Government outlets. The real challenge remains to predict the demand of E-Government services to make the business sustainable (Venables, 2015).

2.2 Government ICT Services

According to (Francis Osanya Wamoto, 2015). e-Government is defined as "the use of ICT such as the wide area network, internet, and mobile computing, by government agencies to transform government operations in order to improve effectiveness, efficiency, service delivery and to promote democracy." Similarly, defines e-government as the use of ICTs to improve public services delivery, and it brings with the promise of greater efficiency and effectiveness in the public sector.

Many studies have been carried out in order to identify critical success factors (CSFs) that influence e-government initiative implementation. The possibility of surveying practices of e-government implementation by applying a limited number of questionnaires to the Ministry of ICT, ICT Authority, the IEBC and the Judiciary was therefore examined. This received good positive responses from many government officers and citizens. The survey was considered timely and valuable because Kenya had implemented e-government system since 2004 and no such study had been performed. To accomplish this, critical success and failure factors were constructed together with their relevant elements to measure the stakeholder's perception of e-government in Kenya (Francis Osanya Wamoto, 2015).

His Excellency, Hon. Uhuru Kenyatta, CGH Former President of the Republic of Kenya, (2017) "The government is entrusted with sustainable development to ensure it uses ICTs to foster economic growth and job creation towards a robust business environment. The ever-present need for ICTs has dramatically morphed into a crosscutting digital tool that my government has steadily taken full advantage of to achieve efficient service delivery to citizens". (Ministry of ICT, Innovation and Youth Affairs, 2022)

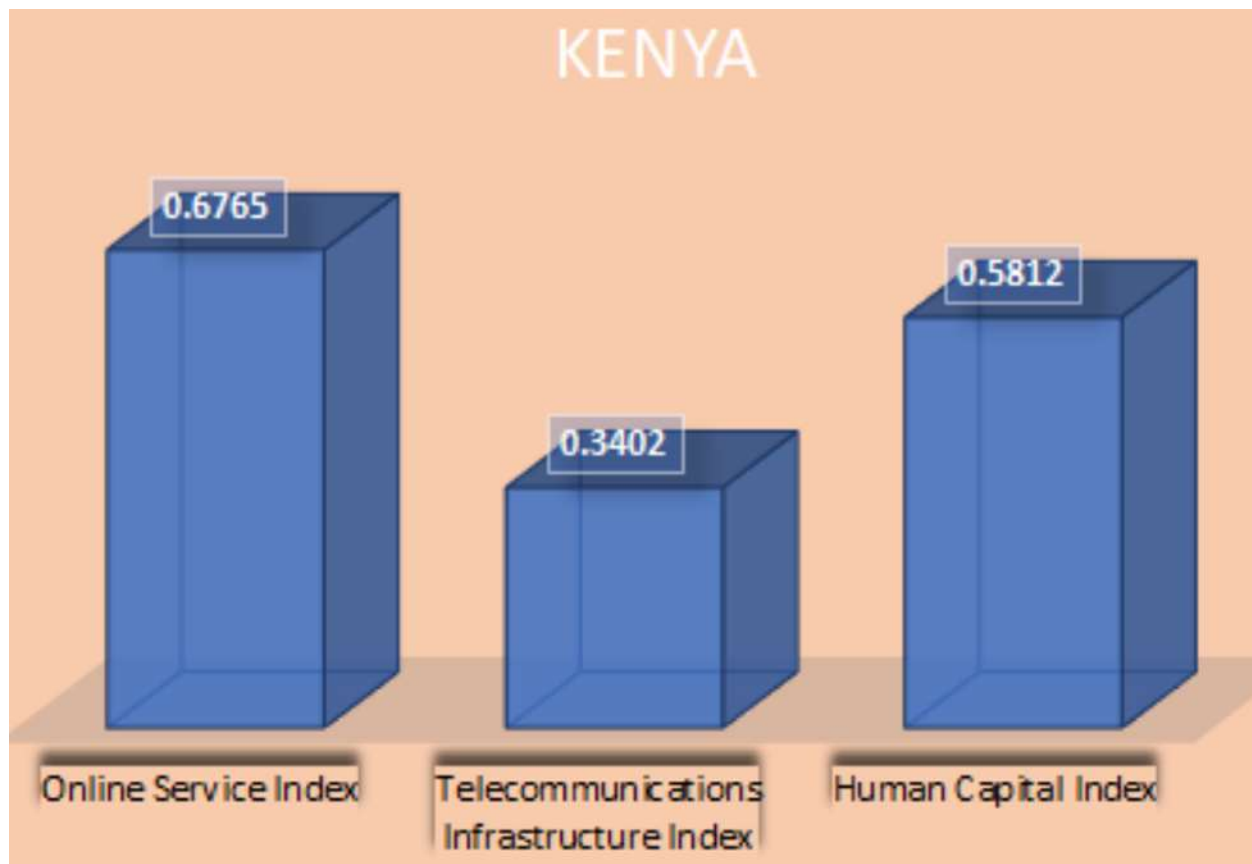
On the digital infrastructure pillar projected under the Kenya National Digital Master Plan 2022 – 2032, the Government envisages to rollout 100,000 Kms of optic fibre installation to all 1,450 wards nationally, digitize Government records and automate all Government systems to maximize the benefits of interoperability under Digital Government Services, products and data management. This plan envisages the building of capacity of 20 million citizens including Special Interest Groups on digital skills, Training of 10,000 officers in Public Service in high-end specialized ICT areas, Training of 300,000 civil servants on digital skills and data protection and

Training of 350,000 teachers on digital skills. (Ministry of ICT, Innovation and Youth Affairs, 2022).

Introduction of Huduma Centers in 2020 as government service stations, exclusively hosts and manages government services. Located in 290 sub-counties across the country and offering services such as applications for national identity cards, passports, driving licenses, business registration, and many more. The program also has a contact center accessible through the call number 020-6900020, Huduma Mashinani Outreaches, and Huduma electronic and mobile platforms. (Huduma Kenya, 2020)

Today, e-government is well known to improve public service delivery. Its adoption results in greater efficiency and effectiveness of government administration. NAIROBI, Kenya June 30 – Kenya's vision of going fully digital in the delivery of government services is rapidly gathering pace with more than 5,000 services now available online. President William Ruto launched the services already onboarded to the eCitizen portal following a presidential directive on Jamuhuri Day last year when the Head of State committed to ease access to government services and eliminate corruption by cutting out physical interactions and brokers. The launch was done concurrently with the unveiling of GavaMkononi, a phone-based app developed to make access to these services easier. The State Department for Internal Security and National Administration is among those that have migrated their operations to the online platform, with four institutions now set to deliver their services digitally. One of the new entrants is the National Authority for the Campaign Against Alcohol and Drug Abuse (NACADA), which will begin licensing importation and exportation of alcoholic drinks online. As part of the government's efforts to improve access to counseling and support services to the grassroots, the Authority is set to accredit rehabilitation centers digitally, with those struggling with addiction also expected to request virtual and in-person training through the same platform. The police Occurrence Book (OB) has also been digitized, a move that will facilitate more efficient and real-time reporting of crimes and security incidents in the country. The digital OB was conceptualized to eliminate manipulation and disappearance of case records since complaints, statements, and details of criminal activities made at police stations will be un-editable (Raymond Omollo, 2023).

The Study below conducted by UN- E-government Survey in 2020 gives a clear analysis on where the country stands as far as infrastructure, online services and human capital index.



e 2.1: UN E-Government Survey 2020, Kenya's Score

Figure 1. UN E-Government Survey 2020, Kenya's Score

2.3 Demand for Government ICT Services

The ever-increasing market growth as well as information awareness across the globe have contributed heavily to the demands for efficient services by citizens. The government of Kenya has adopted the process of reengineering the public service sector to enhance performance and efficiency in service delivery (Korir, Rotich , & Bengat, 2015).

The demand for ICT services is closely tied to the expansion of e-government services. Huduma Centers play a pivotal role in ensuring that citizens can seamlessly transition to and benefit from these digital services.

With the rapid advancements and increased use of Information and Communication Technologies (ICTs), electronic government has gained increased attention from governments, policymakers and practitioners (Krishnan, 2014). Particularly, the advancements in ICTs have fundamentally altered the nature of public administration, to the extent that ICTs now underpin the basic functioning of most public programs and contribute to the most significant innovations in the delivery of public-sector services (Borins, 2001; Dada, 2006; and Holden, 2003). Examples of such services include (but not limited to) tax filing, identity management (including issuance and renewal of identity

cards, driving licenses and passports), application for government jobs, obtaining of birth certificates/marriage licenses, renewal of driver licenses, application for high school grants, registering to vote, and in some cases casting of votes (Baqir and Iyer, 2010).

2.4 Prediction in general

Prediction is the process of making an educated guess or estimation about a future event or outcome based on available information and data. It involves analyzing past patterns and trends, as well as current conditions, to forecast what may happen in the future. (Muhammad Hassan, 2023)

2.5 Prediction for the demand for E_Government services

The integration of machine learning (ML) techniques in the domain of e-government services has witnessed a surge in recent years, driven by the need for efficient resource allocation and improved service delivery. This literature review explores existing ML models applied to predict citizens' demand for government services, providing insights into the evolution of this field and the methodologies employed by various researchers. Prediction for the demand for e-government services involves estimating the future need and utilization of digital government services by citizens (Omar Al-Mushayt, 2019).

a) E-government maturity model

It addresses critical challenges faced by developing countries in sustaining e-government services. Unlike existing models that primarily focus on technological and resource limitations, this novel model introduces a comprehensive approach encompassing five key determinants: a detailed process, streamlined services, agile accessibility, state-of-the-art technology utilization, and trust and awareness. The model emphasizes not only implementation but also incorporates a strategic adoption plan, filling the observed gaps in existing maturity models. Through empirical investigation via case studies and surveys, the model has demonstrated its effectiveness in enhancing the sustainability of e-government services. Both government implementers and service adopters have derived benefits, highlighting its potential to guide developing countries towards successful and enduring e-government initiatives. This model, grounded in a holistic understanding of e-government challenges, provides a valuable reference for related systems, offering insights and strategies for sustainable implementation (Pusp Raj Joshi and Shareeful Islam, 2018).

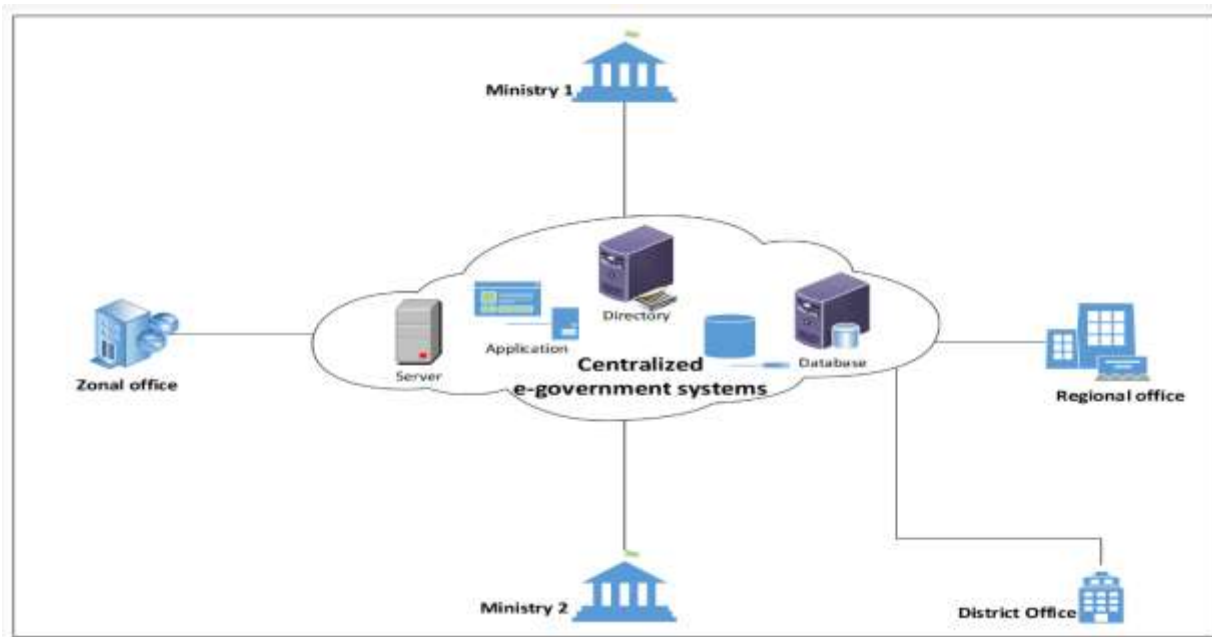


Figure 2. Integration of an e-government system for streamlined e-government services.

2.6 Overview of Machine Learning

Machine learning (ML) is the process of using mathematical models of data to help a computer learn without direct instruction. The adaptability of machine learning makes it a great choice in scenarios where the data is always changing, the nature of the request (Microsoft, 2023). It is a rapidly growing field that has the potential to revolutionize the way we live and work. Machine Learning has drawn a lot of attention. One of the main driving factors of the machine learning hype is related to the fact that it offers a unified framework for introducing intelligent decision-making into many domains Acknowledged as a rapidly evolving field with transformative potential, machine learning extends its influence across diverse applications, notably in image recognition, speech recognition, natural language processing, and predictive analytics (Riccardo Bonetto, Vincent Latzko, 2020)

2.7 Machine Learning and Prediction

In an article written by Sarker, I.H. (2021) the current age of the Fourth Industrial Revolution (4IR or Industry, the digital world has a wealth of data, such as Internet of Things (IoT) data, cybersecurity data, mobile data, business data, social media data, health data, etc. To intelligently analyze these data and develop the corresponding smart and automated applications, the knowledge of artificial intelligence (AI), particularly, machine learning (ML) is the key. Thus, the data management tools and techniques having the capability of extracting insights or useful knowledge from the data in a timely and intelligent way is urgently needed, on which the real-world applications are based.

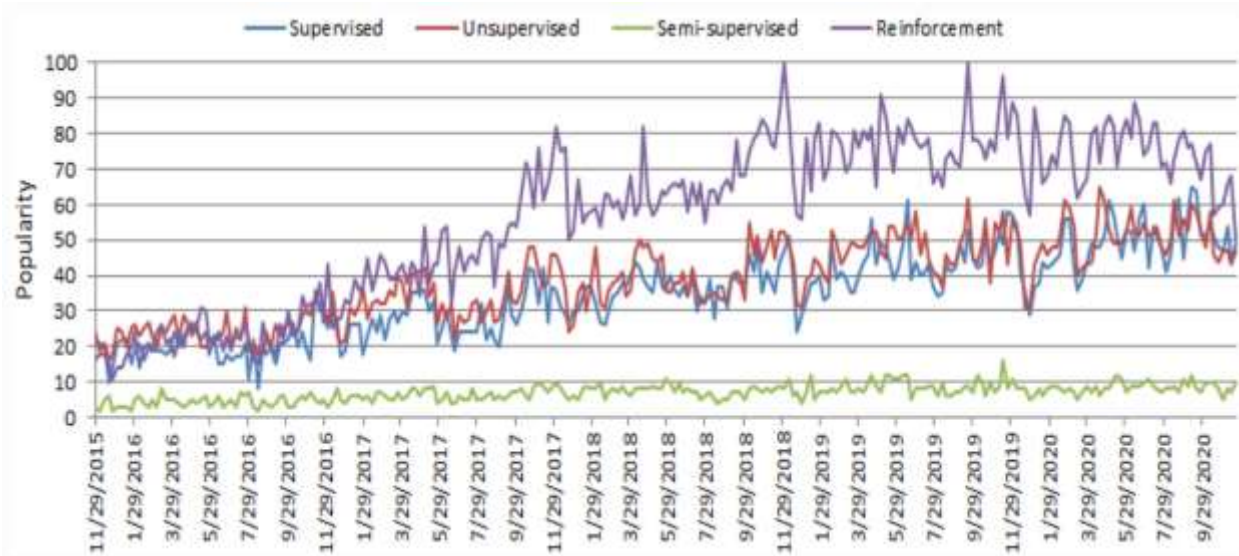


Figure 3. The worldwide popularity score of various types of ML algorithms (supervised, unsupervised, semi-supervised, and reinforcement) in a range of 0 (min) to 100 (max) over time where x-axis represents the timestamp information and y-axis represents the corresponding score

According to Fig. 3, the popularity indication values for these learning types are low in 2015 and are increasing day by day. These statistics motivate us to study on machine learning, which can play an important role in the real-world through Industry 4.0 automation. Artificial intelligence (AI), particularly, machine learning (ML) have grown rapidly in recent years in the context of data analysis and computing that typically allows the applications to function in an intelligent manner. ML usually provides systems with the ability to learn and enhance from experience automatically without being specifically programmed and is generally referred to as the most popular latest technologies in the fourth industrial revolution (Sarker, I.H. 2021).

A general structure of a machine learning-based predictive model is shown below, where the model is trained from historical data in phase 1 and the outcome is generated in phase 2 for the new test data.

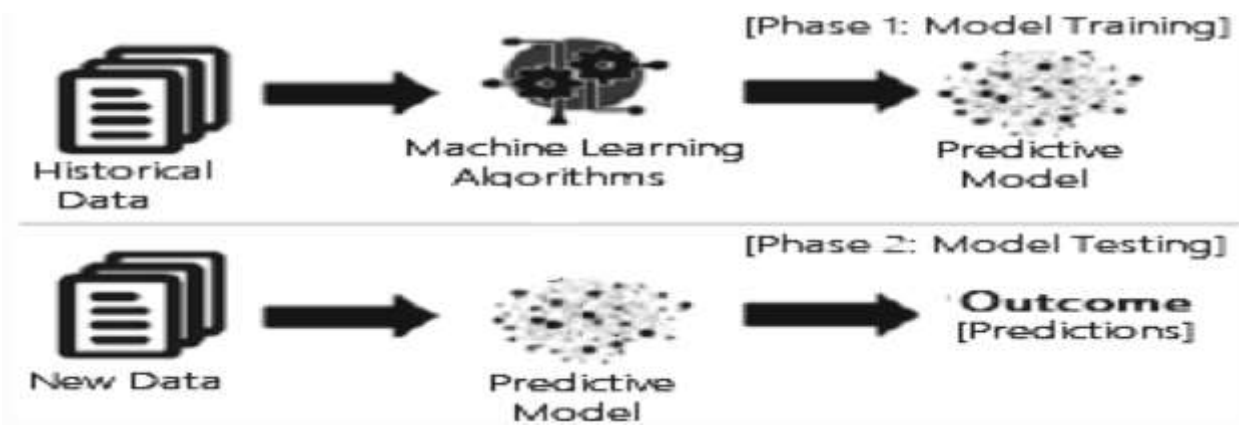


Figure 4. A general structure of a machine learning based predictive model considering both the training and testing phase

2.8 Machine Learning and Prediction of the demand

a) Predicting Kimilili Water Demand Using Artificial Neural Networks

The accurate prediction of time-varying water demand trends and identification of critical demand values are paramount in determining a network's capacity to satisfy essential demand requirements while maintaining economic efficiency. This study was dedicated to predict the water demand of the Kimilili water supply scheme, looking ahead to the year 2030. Operated by Nzoia Water Services Company Limited, the Kimilili water supply scheme faces challenges associated with rapidly increasing water demand, resulting in persistent water supply shortages and unanticipated fluctuations in the system's water production hours. Employing the Artificial Neural Network (ANN) model, this research endeavored to provide insights into predicting water demand for the Kimilili water supply scheme. Notably, the trained model demonstrated commendable performance, boasting a coefficient of determination (R^2) of 0.999972988, underlining its efficacy in predicting water demand patterns for this particular scheme (Shilehwa, Celsus Murenjekha Makhanu, Sibilike Khamala Khaemba, Alex Wabwoba, 2019).

b) Enhancing Employment prediction in Kenya's Logistics Sector

The study undertaken to estimate employment demand within Kenya's logistics sector presents a thorough investigation into alternative methodologies. Emphasizing the imperative need to grasp the current state of the country's industry, the research critically reviews several established models for employment forecasting, including those utilized in the USA (Bureau of Labor Statistics), Netherlands (ROA), UK (IER - Institute for Employment Research), and Germany (IAB). A central proposal of the study advocates for a time-lag approach in forecasting employer demand within the logistics domain, contending that this method offers a more realistic anticipation of future needs based on the temporal progression of industry output. The empirical findings of the study, anchored in the time-lag approach and employing industry output as a key metric, reveal significant insights. For instance, considering an industry output of 733,080 KSH million in 2015 and 970,640 in 2020, the projected demand for wage employment in the logistics sector is estimated at 95,860 in 2015 and 104,329 in 2020, surpassing the 2012 figure of about 89,600. The study posits that these results offer a more rational and updated perspective on employment demand compared to prior research endeavors.

2.9 The Gap

Table 2. The gap in the current systems

Project Name	Identified Gap	Proposed System Solution
Predicting Kimilili Water Demand Using Artificial Neural Networks	Lack of data on water consumption patterns and climate change impact.	Comprehensive data collection, including historical usage patterns and environmental factors. Adaptable predictive modeling techniques integrate evolving environmental conditions.
Enhancing Employment Prediction in Kenya's Logistics Sector	Limited consideration of Kenya's logistics sector nuances in existing forecasting models.	In-depth analysis specific to Kenya's logistics sector, incorporating factors beyond industry output. Engage local industry experts and stakeholders for insights. Evaluate various forecasting methodologies, including time-lag approaches.

2.9 Proposed system architecture

In addressing the deficiencies observed in current demand prediction models, the proposed model harness historical data spanning the last three years, the data is obtained from the Huduma Kenya database. This strategic utilization of an extensive dataset enhances the likelihood of crafting an optimal predictive model.

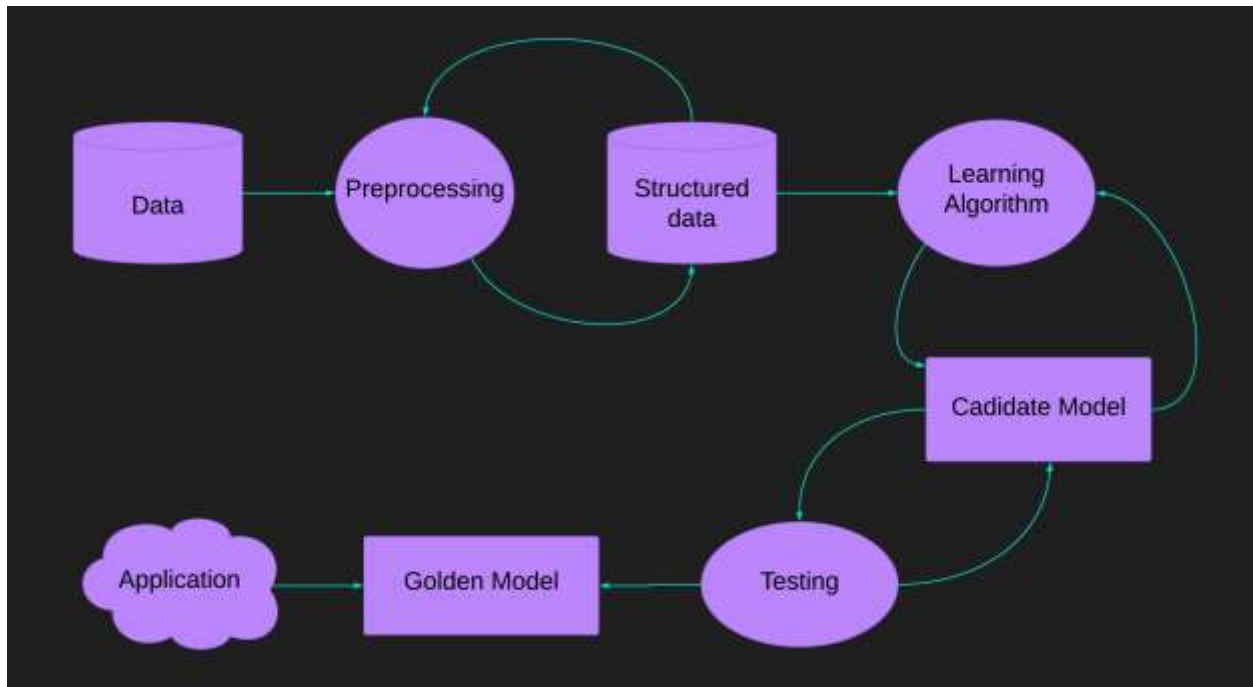


Figure 5. Proposed System Architecture

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This project will leverage the Agile methodology. The rationale behind this selection is grounded in the following considerations,

1. Agile allows for flexibility in responding to changing requirements. In a predictive modeling project, initial assumptions and requirements may evolve as insights are gained. Agile accommodates these changes seamlessly.
2. Delivery of working software at the end of each iteration, aligns well with the iterative nature of model development, allowing for the continuous delivery of incremental improvements.
3. Agile encourages regular collaboration with stakeholders, including end-users. Predictive models are often refined based on user feedback and changing business needs
4. Model development often involves iterations of training and refining based on results. Agile's iterative nature aligns well with the iterative training and evaluation cycles required for developing an accurate predictive model.

To facilitate the development of the model through multi-linear regression in adherence to Agile methodology, the subsequent steps will be followed,

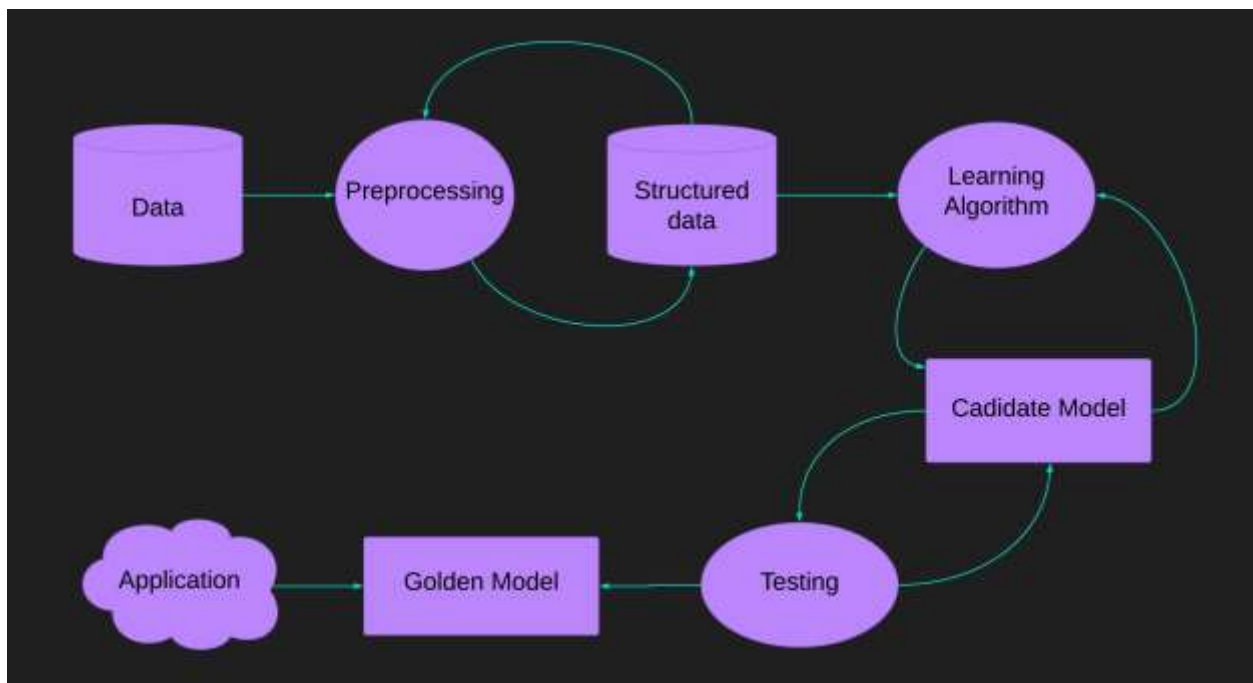


Figure. 6. Demand prediction development process.

Step 1: Data Collection

Data collection involved extracting data from the Huduma center database system, encompassing appointment records spanning the last three years of Huduma center operations.

Step 2: Data Preprocessing

Data preprocessing was imperative for ensuring data quality. This step involved comprehensive cleaning, handling of missing values, and transforming the raw data into a standardized format conducive to model utilization.

Step 3: Feature Selection

Identification of key features influencing service demand was critical. Leveraging correlation analysis, I pinpointed independent variables most relevant to the dependent variable, thereby streamlining the model's predictive accuracy.

Step 4: Model Selection

At this stage, the selection of the most suitable machine learning model was carried out, ensuring alignment with the specific requirements of demand prediction. This decision-making process involved a thoughtful evaluation of the unique characteristics and complexities inherent in the demand prediction task to identify the model that best addresses these specific needs

Step 5: Model Training

To optimize model performance, the dataset will be partitioned into training and testing subsets. The model will then be trained on the training set, fine-tuning parameters for optimal fit.

Step 6: Model Prediction and Evaluation:

The model's predictive capabilities were evaluated rigorously using a distinct test set. The evaluation process will focus on accuracy metrics to ascertain that the model effectively generalizes beyond the training data without overfitting.

Step 7: Testing and Iteration

Continuous testing and iteration are paramount. Agile principles dictate ongoing testing throughout development, ensuring that each increment is not only functional but aligns with evolving user needs. The testing phase will conclude with a thorough evaluation, paving the way for potential iterations and enhancements.

Step 8: Model Deployment:

Once the model has been trained and evaluated, it will be deployed to either google cloud to make predictions on new data.

3.2. CONSTRAINTS

Security and Privacy: - since the system involves handling of peoples' sensitive data, it will require extra measures to safeguard the integrity and privacy.

Regulatory Requirements: - compliance with data protection regulations, ethical considerations in data usage, and any other legal obligations relevant to the implementation of the predictive model within the public service domain is required, and this may pose some delay in obtaining the data.

Hardware Limitations: - Constraints related to hardware, such as processing power and storage due to the size of the data, may impact the system's capabilities.

3.3. RESOURCES

3.3.1 Hardware resources

This refers to the physical components of the computer system required to develop and implement the demand prediction system. The hardware resources required are;

1. Computing Power: Sufficient CPU power to handle data processing and model training.
2. GPU (Graphics Processing Unit) support for accelerated machine learning tasks
3. Memory (RAM): Adequate RAM for efficient handling of datasets and model training is required at least 8GB RAM and above is required.
4. Networking: Stable and high-speed internet connection for data retrieval, model updates, and potential integration with external data sources.

3.3.2 Software resources

This refers to the software components required to develop and implement the demand prediction system. The software resources required are;

1. Programming language: will utilize the Python programming language as the primary coding platform. The project will specifically leverage the scikit-learn library for implementing the multi-linear regression algorithm.
2. Integrated Development Environment (IDE): Jupyter Notebook for interactive development and data exploration will be used for this project.
3. Hugging Face : Integration with Hugging Face for hosting the model

3.3.3 People resources

This refers to the human resources required for developing and implementing the demand prediction system. The people resources required are;

1. Data Scientists: Individuals with expertise in data analysis, feature engineering, and model training.

2. Python Developer: Developers proficient in Python programming for implementing data processing and building machine learning models.
3. Machine Learning Engineer: Specialists who can design and implement machine learning algorithms and models.
4. System Administrators: Professionals responsible for managing the hardware infrastructure, ensuring adequate resources, and maintaining system stability.

3.4 Feasibility Study

This examination is conducted to assess the viability of the system and its requirements within specified constraints such as time, financial resources, operability, and more. In the context of this prediction system, a thorough analysis will be undertaken considering the following parametric constraints.

3.4.1 Operational Feasibility

This measures if the proposed system will solve the problem defined and meet the objectives stated, it also measures the users' reaction/ feedback about the system.

3.4.2 Resource Feasibility

Assessment of the computing resources, personnel and physical facilities. Predictive system was feasible resource-wise since its personnel are available and skilled to complete the project on time.

3.4.3 Economic Feasibility

The measure of the project's costs and revenues, whether it is affordable for the project's completion. Predictive system cost benefit analysis is as follows: -

- i) Onetime costs of the system e.g., time, and personnel beat the recurring costs since there are no licensing agreements nor programming support.
- ii) There are no purchases of the tools to develop the system. Most tools are open source and free.
- iii) The returns from the system's operation will tend to recover for the resources used to build the system.

3.4.4 Technical Feasibility

The prediction system is technically feasible since it uses technology that has been used to develop applications and will continue being used in years to come e.g. Python and SQL.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGNS

4.1 Introduction

In this chapter proper planning of the development of the system through understanding and specifying in detail what this system should do and how the components of the system should be implemented and work together was done.

4.2 System Analysis

4.2.1 Requirements Gathering

Below are different methods utilized in gathering the requirements.

4.2.1.1 Review of Existing documents

A comprehensive review of literature pertaining to e-government services delivery was conducted to gain a deep understanding of the domain in which this system operates. Various sources were explored, including scholarly works by renowned authors such as Al-Mushayt (2019) in "Automating E-Government Services with Artificial Intelligence," Joshi and Islam (2018) in "E-Government Maturity Model for Sustainable E-Government Services," and Wahida (2016) in "Service Delivery in the Public Sector," among others cited in the reference section of this document.

Through this extensive review, insights were gathered into the landscape of government services offered electronically, providing valuable context for the development of the system at hand.

4.2.1.2 Visit to government Institutions

As part of the requirements gathering process, conducted on-site visits to government institutions providing E-Government Services. Specific visit at Huduma Center Makandara and engagement with the Huduma Kenya ICT department was made to gain firsthand insight into the entire process that citizens undergo when accessing essential services. This immersive experience provided invaluable understanding of the journey citizens undertake to obtain eagerly awaited services, as well as the challenges encountered along the way such as long queues, among others.

During these visits, it was an opportunity to consult with one Mr. Fredric Kimingi from the ICT department, who offered detailed explanations on how various information systems have been instrumental in addressing the needs of Huduma Center users. The visits yield access to datasets containing information about the diverse range of services available at Huduma Center.

Statistical Data Review

The dataset utilized in this project originates from the Huduma Center database, which is a government initiative designed to facilitate citizens' access to an array of government services. Huduma Center stands as a central hub, offering essential government services tailored to citizens' needs, thereby serving as a comprehensive repository of pertinent data for this project. The

deliberate selection of the Huduma Center database as the primary data source underscores the project's commitment to relevance and practicality, ensuring that the prediction system is finely attuned to the real-world requirements of individuals seeking government services. Leveraging data from Huduma Center aligns with the project's objective of enhancing service delivery and efficiency within government agencies through predictive analytics.

4.3 Requirements Specifications

Following the comprehensive process of requirements gathering, the next crucial phase, to document and articulate these gathered insights into concrete and actionable Requirements Specifications. This phase serves as a pivotal bridge between understanding user needs and translating them into specific guidelines for system development

4.3.1 Functional requirements

The system should

1. Authenticate users before accessing the system,
2. Predict quarterly demand for a specific E-government service,
3. User action verification. e.g the user cannot predict service until they enter the required inputs
4. Visualization report of the predicted demand.

4.3.2 Non-Functional requirements

1. The system should be capable of processing large volumes of data efficiently and providing timely predictions to support decision-making processes.
2. The solution must be scalable to accommodate increasing data volumes and user demands over time without compromising performance or accuracy.
3. The system should be robust and resilient, with mechanisms in place to handle errors, failures, and disruptions gracefully, ensuring continuous availability and data integrity.

4.3 System Design

4.3.1 Architectural Design

This step involves translating the requirements identified into a comprehensive architecture and design that fulfils the project objectives. The processes involved in designing the architecture and infrastructure necessary to support the development of Predictive model.

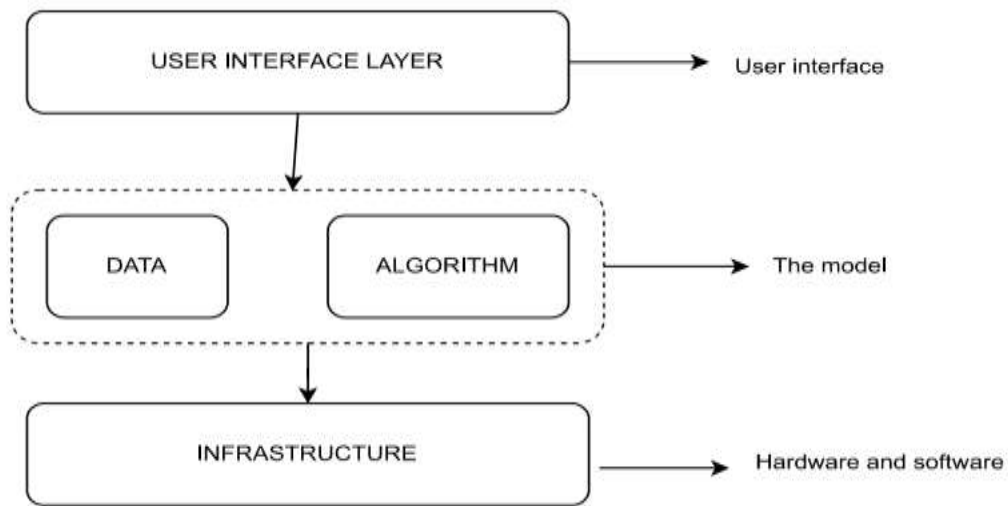


Figure 7: Top-level design in the context of the whole system

4.3.2 Data Design

4.3.2.1 Entity Relationship Diagram (ERD)

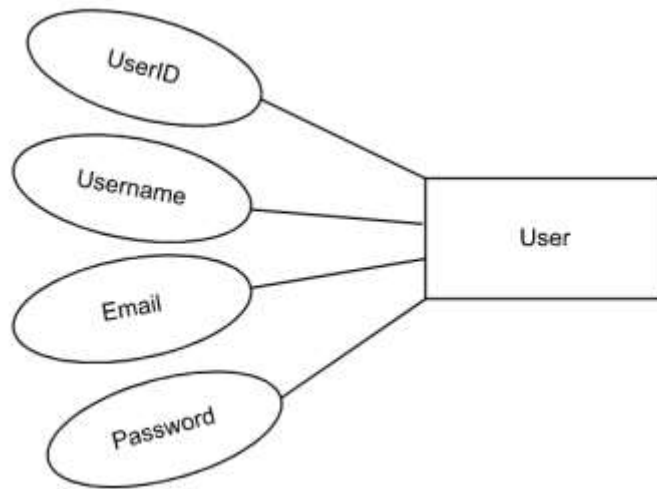


Figure 8: Data model

4.3.3 Process Design

This step defines the processes within the demand prediction model, including the components, modules, and interactions required to support data ingestion, pre-processing, modelling, visualization, and integration with existing systems.

4.3.2.1 Data Flow diagrams

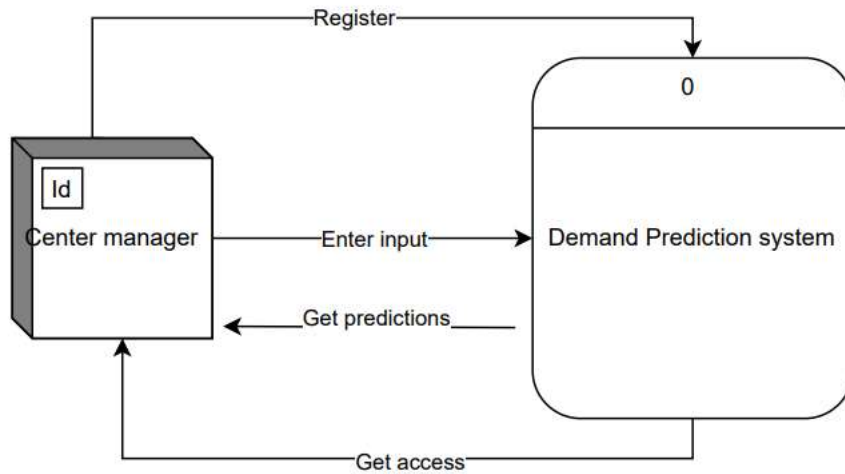


Figure.9 Context Diagram

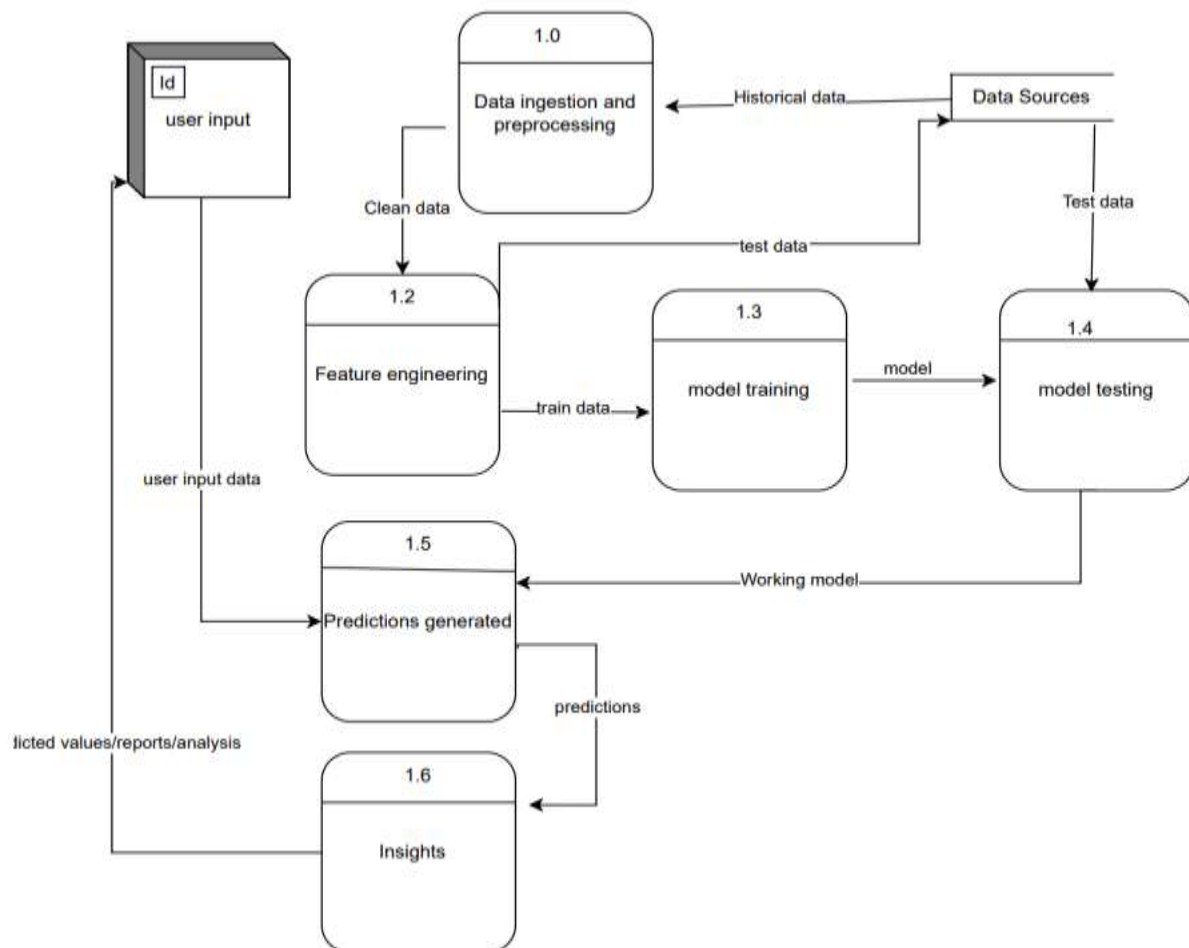


Figure.10 Level 0 Diagram

4.3.3 Database diagram

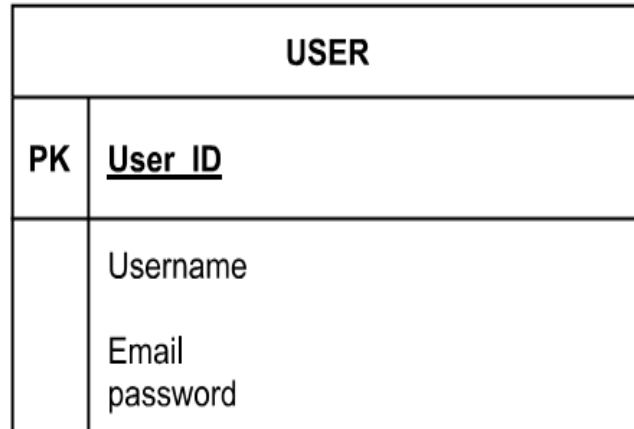


Figure.11 Database ERD

Use Case design

Use case 1

Use case name: Credential Authentication.

Entry condition.

The use case starts when a user launches the system. The user is prompted to enter username and password to get access to the system. After a successful login the user is automatically redirected to the main dashboard of the application.

Exit condition.

The use case terminates when the user closes the dashboard.

Use case 2

Use case name: Upload Data.

Entry condition.

This use case begins when the user clicks ‘Upload Data’ button on the dashboard. The file explorer is launched and user is prompted to upload file if the file is of type comma separated version or xlsx version the data is uploaded and preprocessed which upon completion the user gets a ‘data preprocessed successfully’, message.

Exit condition.

The use case terminates when the user closes the dashboard.

Use case diagram

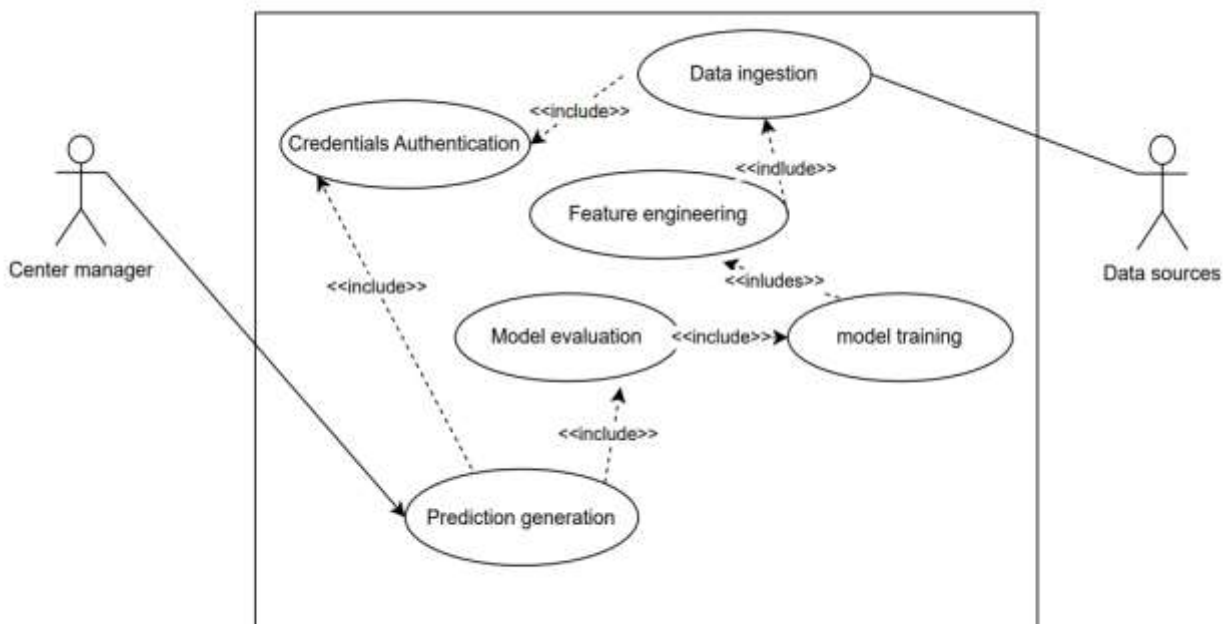


Figure 12: Use case diagram

Flow chart

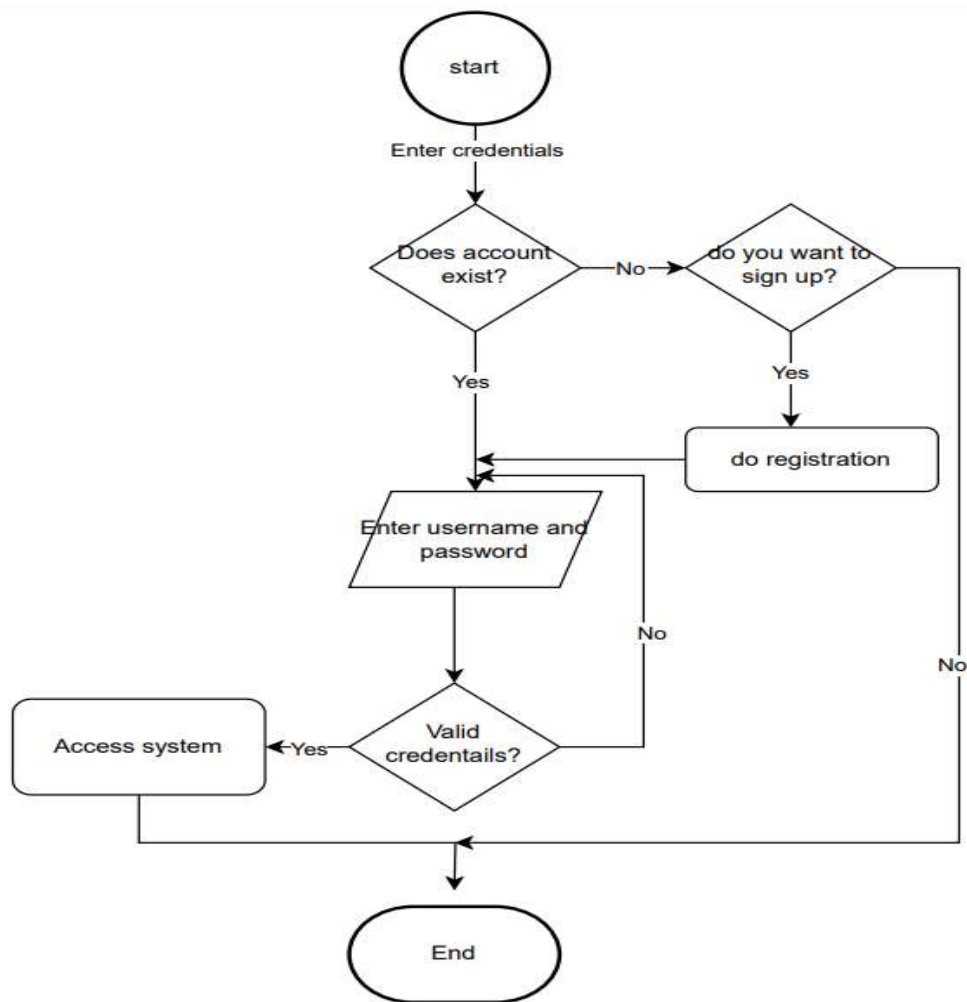


Figure 13: Flow Chart diagram

4.3.4 User Interface Design

Mock-ups and wireframes were developed to visualize user interface of the demand prediction model. .

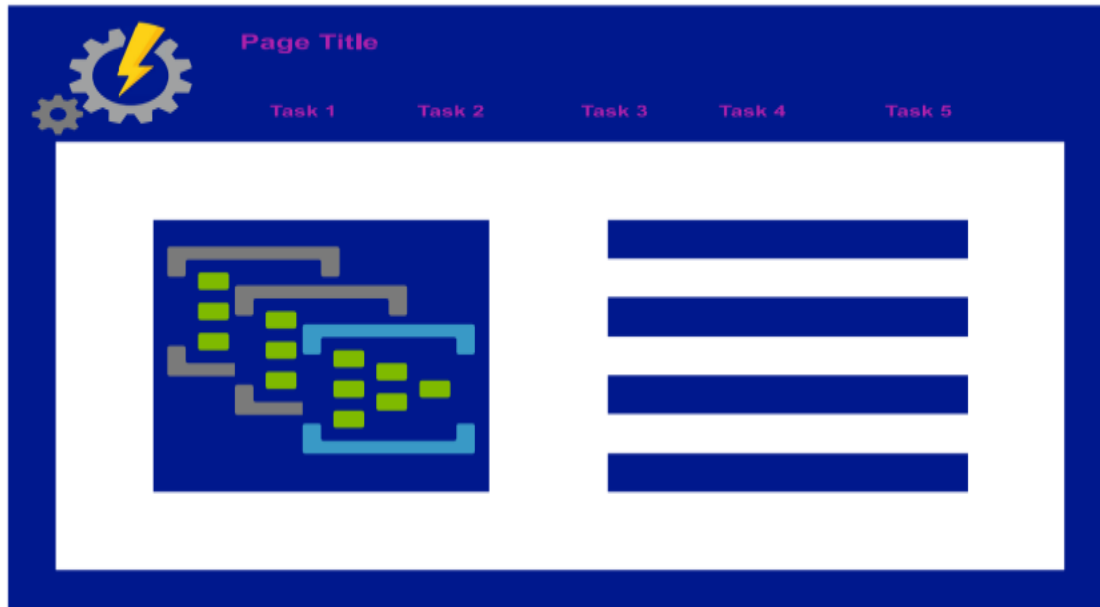


Figure 14. Main dashboard of the prediction system

4.4 Summary of the chapter

The System Analysis and Design chapter serves as a crucial phase in the development process, focusing on translating the project's conceptual design and theoretical framework into an operational solution. It begins with an introduction outlining the objectives of the chapter and proceeds to delve into various aspects of system analysis and design.

Several key sections, including requirements gathering, specification of functional and non-functional requirements, and system design. Within the system design section, architectural design, data design, process design, database design, flowchart, and user interface design are covered in detail.

Throughout the chapter, a structured approach is adopted to ensure the thorough analysis and design of the predictive model system. This involves identifying and documenting user requirements, defining system specifications, and designing a robust system architecture to support the project's objectives.

CHAPTER 5: SYSTEM IMPLEMENTATION AND TESTING

5.1 Introduction

In this chapter, the implementation phase of the e-government service prediction system begins, marking a significant milestone in the development lifecycle. The primary objective being to translate the conceptual design and theoretical framework established in previous chapters into a tangible, operational solution. The implementation process is planned and executed, following the principles of the Agile methodology to ensure efficiency, flexibility, and adaptability.

5.2 Analysis of the Dataset

The dataset contains the citizen data entered during booking for appointments at huduma center, and it contains the following features;



	type	dob	placeofbirth	gender	residencecounty	quarter	servicename	booked_date	booked_time	stationname	callsource
0	online-booking	1988-02-26 00:00:00	KAKAMEGA	M	47	20202021Q1	KRA - KRA Services	2020-08-19	2020-08-19 13:00:00	GPO HUDUMA CENTRE	SBA
1	online-booking	1987-02-22 00:00:00	KIRINYAGA	M	47	20202021Q1	NRB - Application for Duplicate ID	2020-08-19	2020-08-19 11:00:00	GPO HUDUMA CENTRE	SBA
2	online-booking	1995-08-25 00:00:00	DAOGORETTI	F	47	20202021Q1	NRB - Collection of ID	2020-08-17	2020-08-17 08:00:00	GPO HUDUMA CENTRE	SBA

Figure 15: Data

Type: The type of booking method used by citizen to book for service.

Service Name: The service that the visitor seeks at the Huduma center.

Date of Birth (DOB): The date of birth of the visitor.

Place of Birth: The birthplace of the visitor.

Gender: The gender of the visitor.

Residence County: The county of residence of the visitor.

Quarter: The time period (quarter) of the visit.

Booked Date: The date when the visit was booked.

Booked Time: The time when the visit was booked.

Station Name: The name of the Huduma center station.

Call Source: The source from which the visit was initiated.

5.3 Data Preprocessing

Aimed at preparing raw data for model training and analysis, in this section, we delve into the fundamental data preparation activities essential for building robust and accurate machine learning

algorithms. From handling missing data to encoding categorical variables and scaling features, each preprocessing task plays an important role in optimizing the performance of predictive this model. Steps involved in data preprocessing are as follows;

5.2.1 Data cleaning

This is the initial step where identification and addressing of any inconsistencies, errors, missing values in the data are taken care of, to ensure the dataset is clean and free from any anomalies. This lays the foundation for accurate and reliable model training and analysis.

Evidence!

3.1 Data Cleaning

```
1 data.drop(columns=["callsource"], inplace = True)
2
3 problematic_rows = data['dob'] == '0000-00-00'
4 problematic_data = data[problematic_rows]
5 data = data[~problematic_rows]
6
7 problematic_rows = data['dob'] == '\\N'
8 problematic_data = data[problematic_rows]
9 data = data[~problematic_rows]
10
11 # Filter out rows with 'UNKNOWN' values in the 'placeofbirth' column
12 unknown_rows = data['placeofbirth'] == 'UNKNOWN'
13 data = data[~unknown_rows]
```

Figure 16: Data cleaning

5.2.2 Feature engineering

This step focused on transforming raw data into meaningful features that are to be used as inputs for the model. Activities done here were creating new features, transforming existing features, or selecting relevant features that are most predictive of the target variable. Feature engineering aims to enhance the performance of machine learning models by providing them with relevant and informative inputs that capture the underlying patterns and relationships in the data.

Evidence!

```

1 data['moving_avg'] = data.groupby('servicename')['demand'].transform(lambda x: x.rolling(4, min_periods=1).mean())
2
3
4 # Calculate year-over-year change in demand
5 data['yoy_change'] = data.groupby(['servicename', 'quarter'])['demand'].transform(lambda x: x.pct_change(4))
6
7 # Calculate quarter-over-quarter change in demand
8 data['qoq_change'] = data.groupby(['servicename'])['demand'].transform(lambda x: x.diff())
9

```

Figure 17: Feature engineering

5.2.3 Categorical data transformation

In this step encoding of categorical variables into numerical representations that can be processed by machine learning algorithms was done. This step is essential because most machine learning models require numerical inputs. Techniques utilized in transforming the categorical data was label encoding and frequency encoding.

Evidence!

```

# Initialize LabelEncoders for categorical columns
label_encoders = {}
categorical_columns = ['quarter', 'servicename']

for column in categorical_columns:
    label_encoders[column] = LabelEncoder()
    data[column] = label_encoders[column].fit_transform(data[column])

# Save the encoders for future use
joblib.dump(label_encoders, 'label_encoders.pkl')

```

Figure 18: converting categorical values

Transformed data



1 data.head()

53]:

	year	quarter	servicename	demand	moving_avg	yoy_change	qoq_change	demand_lag1	demand_lag2
2	0	1	3	1518	1518.0	-23.16391	-3.143514	179.0	77.0
3	0	1	6	58	58.0	-23.16391	-3.143514	1518.0	179.0
4	0	1	7	1315	1315.0	-23.16391	-3.143514	58.0	1518.0
5	0	1	8	40	40.0	-23.16391	-3.143514	1315.0	58.0
6	0	1	9	2	2.0	-23.16391	-3.143514	40.0	1315.0

Figure 19: Transformed data

5.4 Feature selection

The most relevant and informative features are identified from the dataset while discarding redundant or irrelevant features. This step helps reduce the dimensionality of the dataset, improve model performance, and enhance interpretability by focusing on the most important predictors.

Evidence!

```
# Define features and target variable
X = X_encoded[['servicename_encoded', 'stationname_encoded', 'quarter_encoded', 'year_encoded']]
y = X_encoded['demand_bin_encoded']
```

Figure 20: Feature selection

5.5 Model training

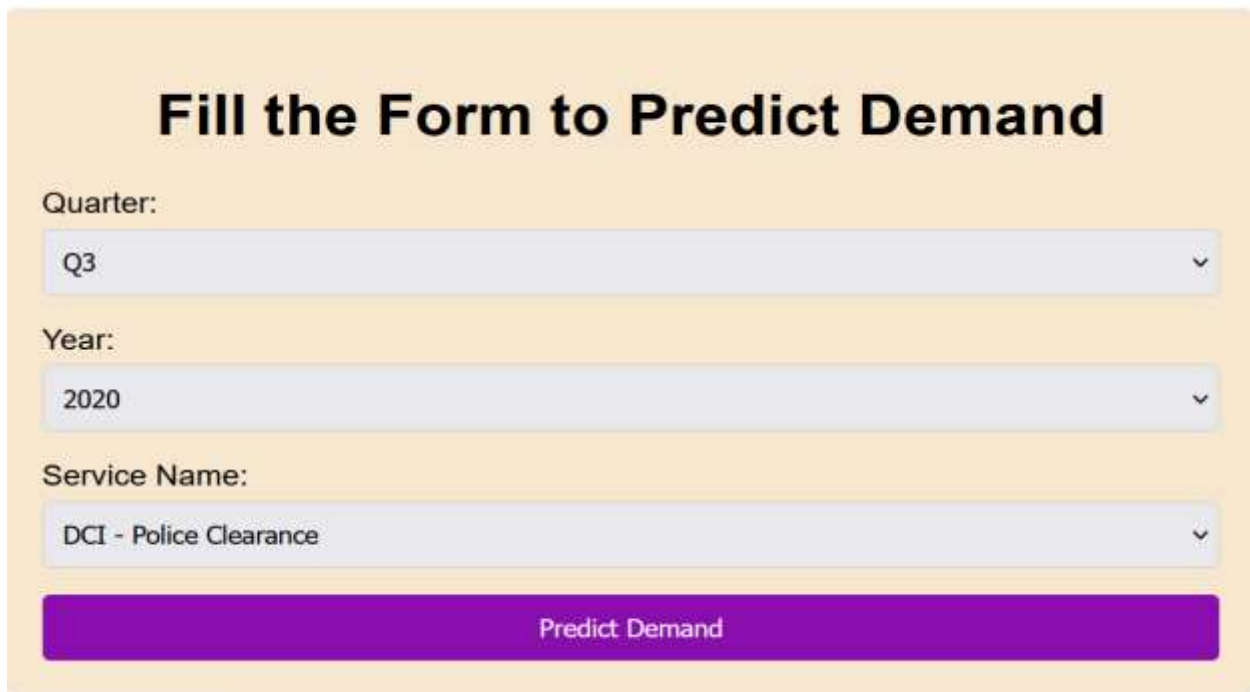
During this step, the algorithm is trained on a labeled dataset, where the input features are used to predict the target variable. The model is optimized by adjusting its parameters to minimize a chosen loss function, such as mean squared error or cross-entropy loss.

Evidence!

```
22 # Train the model
23 model = RandomForestRegressor(n_estimators=100, random_state=42)
24 model.fit(X_train, y_train)
```

Figure 21: Model Training

5.6 User interface Implementation



Fill the Form to Predict Demand

Quarter:
Q3

Year:
2020

Service Name:
DCI - Police Clearance

Predict Demand

Figure22: Demand prediction User Interface

5.7 Prediction

The trained model is then evaluated using validation data to assess its performance and generalization ability before being deployed for making predictions on new, unseen data.



Figure 23: Predicted demand

5.8 Testing and Evaluation

5.8.1 Quantitative Evaluation

To assess the performance of the demand prediction model, a quantitative evaluation was conducted using the following metrics:

```

12]: 1 # Calculate the mean squared error and R-squared score
      2 mse = mean_squared_error(y_test, y_pred)
      3 r2 = r2_score(y_test, y_pred)
      4 mae = mean_absolute_error(y_test, y_pred)
      5
      6 print(f"R-squared Score: {r2:.2f}")

```

R-squared Score: 0.90

Figure 24: Accuracy of the prediction

R-squared (R^2): This metric measures the proportion of variance in the dependent variable (quarterly demand for government services) that can be explained by the independent variables in the model. An R-squared value of 0.90 (90%) indicates that the model accounts for 90% of the variance in the quarterly demand, effectively capturing the fluctuations in demand.

5.8.2 Domain Expert Evaluation

To assess the performance and real-world applicability of the demand prediction system, a structured evaluation was conducted with the domain expert from the Makadara Huduma Center. The evaluation consisted of a questionnaire designed to gather both quantitative ratings and qualitative feedback from the expert's perspective based on the output of the system compared to the real world.

5.8.2.1 Predicted Demand for DCI Police Clearance demand

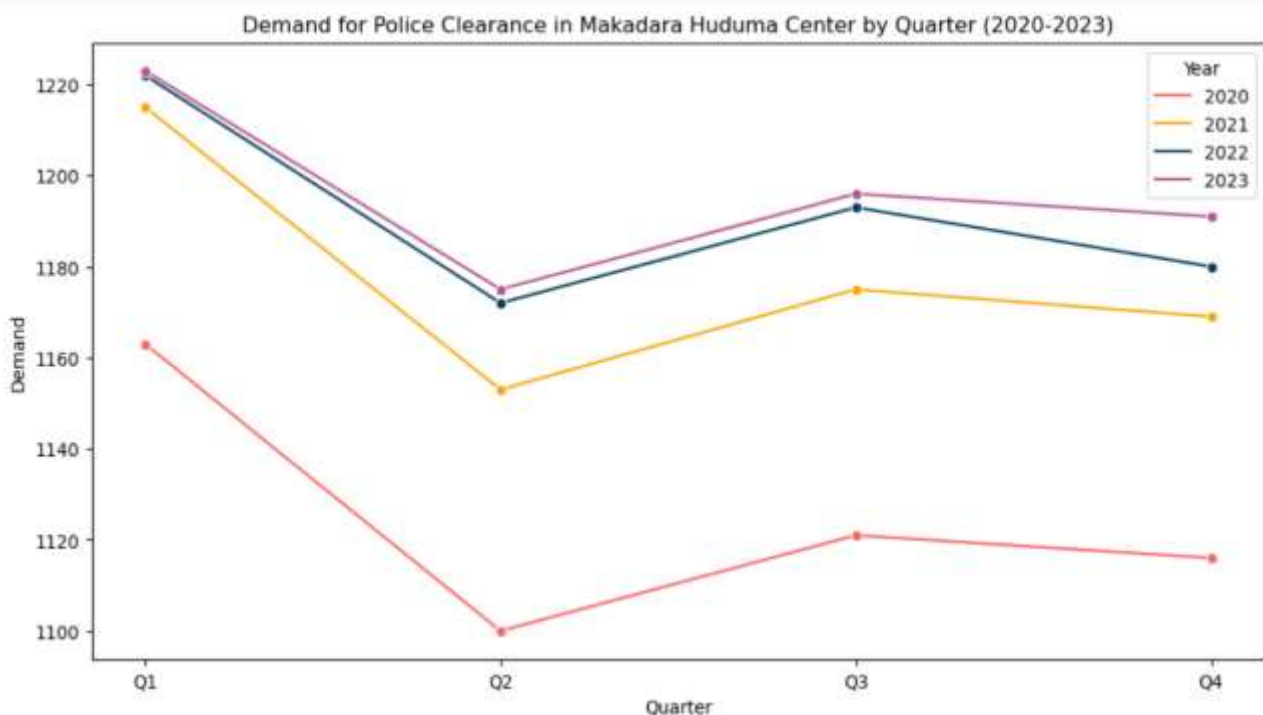


Figure 25: Police Clearance Demand visualized

5.8.2.2 Predicted Demand for ID Collection service

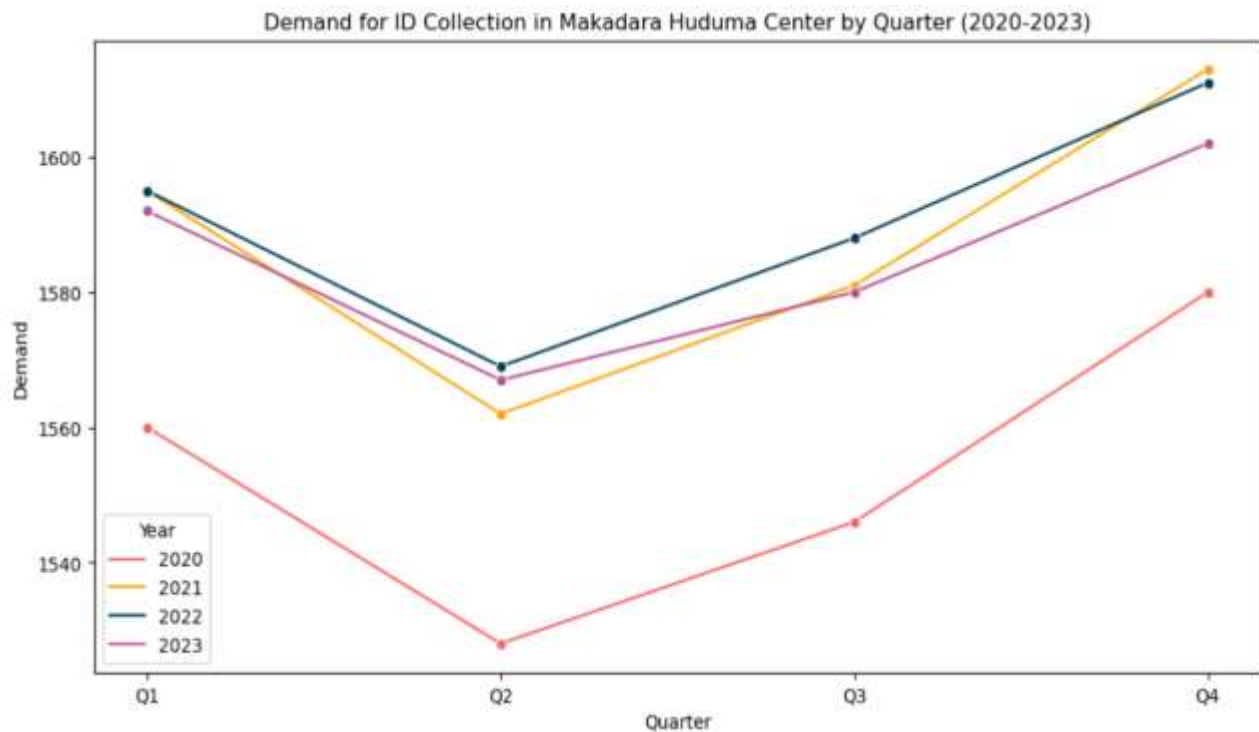


Figure 26: ID Collection Demand visualized

Questionnaire

Part A: Open-Ended Questions

1. Do the predictions align with the real-world patterns and trends you have observed in Huduma service demand? **Yes, they do.**
2. Are there any specific service categories or time periods where the predictions seem to deviate significantly from your expectations? **The deviations are consistent for all periods at 83% accuracy level. The model will require further training for higher accuracy on its predictions.**
3. What are the strengths and weaknesses of the modelling approach used, from your perspective? **Random forests are a powerful and versatile machine learning algorithm which builds an ensemble of decision trees and aggregates their predictions to make more accurate and robust predictions. Its strengths lie in its accuracy, robustness to overfitting, and ability to handle high-dimensional data, making it suitable for a wide range of regression tasks. However, they also come with some weaknesses, including computational complexity, interpretability issues, and potential challenges with high-dimensional and noisy data. It is good to balance out the pros and cons to have a better outcome.**

4. Are there any additional factors or variables that the system should consider to improve the accuracy and applicability of the predictions? **The independent variables used as sufficient for this prediction.**

Part B: Rating Scale Questions

1. Based on your experience, on a scale of 1 to 5 (where 1 is "not accurate at all" and 5 is "extremely accurate"), how accurate do you find the demand predictions generated by the system? (Please provide a rating and any additional comments or examples to support your rating.) **I will give it a 4 rating. The accuracy of the prediction is at 83%. The algorithm needs several hundreds of rounds of training before it can give prediction with an error margin of at least 5%**

How useful and actionable do you find the demand predictions for resource planning, staffing, or operational decision-making? (Please provide a rating on a scale of 1 to 5, where 1 is "not useful at all" and 5 is "extremely useful and actionable," along with any additional comments or feedback.) **I will a rating of 5. This is extremely important to ensure enhanced service delivery.**

2. On a scale of 1 to 5 (where 1 is "poor" and 5 is "excellent"), how would you rate the overall real-world applicability and usefulness of the demand prediction system for Huduma services? (Please provide a rating and any additional comments or suggestions for improvement.) **I would give it a rating of 5.**

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 Summary of achievement

The primary objective of the system was to predict quarterly demand for specific E-government services. This objective, defined under the functional requirements, has been successfully achieved. The system leverages historical data and advanced machine learning algorithms to generate accurate demand predictions for various services on a quarterly basis. This functionality enables government agencies to better allocate resources, improve service delivery, and enhance user satisfaction. The successful implementation of this predictive capability demonstrates that the system has met its core objective.

6.2 Challenges faced during development

During the development of the project, several challenges were encountered. The system was initially designed to predict demand for E-government services across all Huduma centers in Kenya. However, due to the large volume of data required for this comprehensive scope, the system faced significant performance issues, as limited resources in terms of computer memory, processor speeds, and storage capacity were insufficient to manage and process the large volumes of data involved. This meant narrowing the scope to a smaller subset of data, specifically to predict the demand at one Huduma center which is Makadara Huduma Center.

6.3 Future Work and Improvements

The successful completion of the system and its achievement of the set objective opens up opportunities for future work and improvements. This could involve developing a more advanced demand prediction system capable of handling larger datasets and providing comprehensive predictions for multiple locations by leveraging scalable cloud computing resources, incorporating more sophisticated machine learning algorithms, and integrating real-time data processing capabilities. This could offer more accurate and extensive demand predictions, aiding in better resource allocation and service delivery across all Huduma centers in the Kenya.

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18. Image by Freepik

APPENDICES

APPENDIX A: User Manual

The user first encounters the main page. On this page, there is a "Login" button that, when clicked, prompts the user to enter their username and password to access the application. If the user does not have an account, they can click on the "Create Account" button to register for a new account. This simple and intuitive interface ensures easy navigation and access for all users.



Figure 27: Login page

Appendix B: Login Page

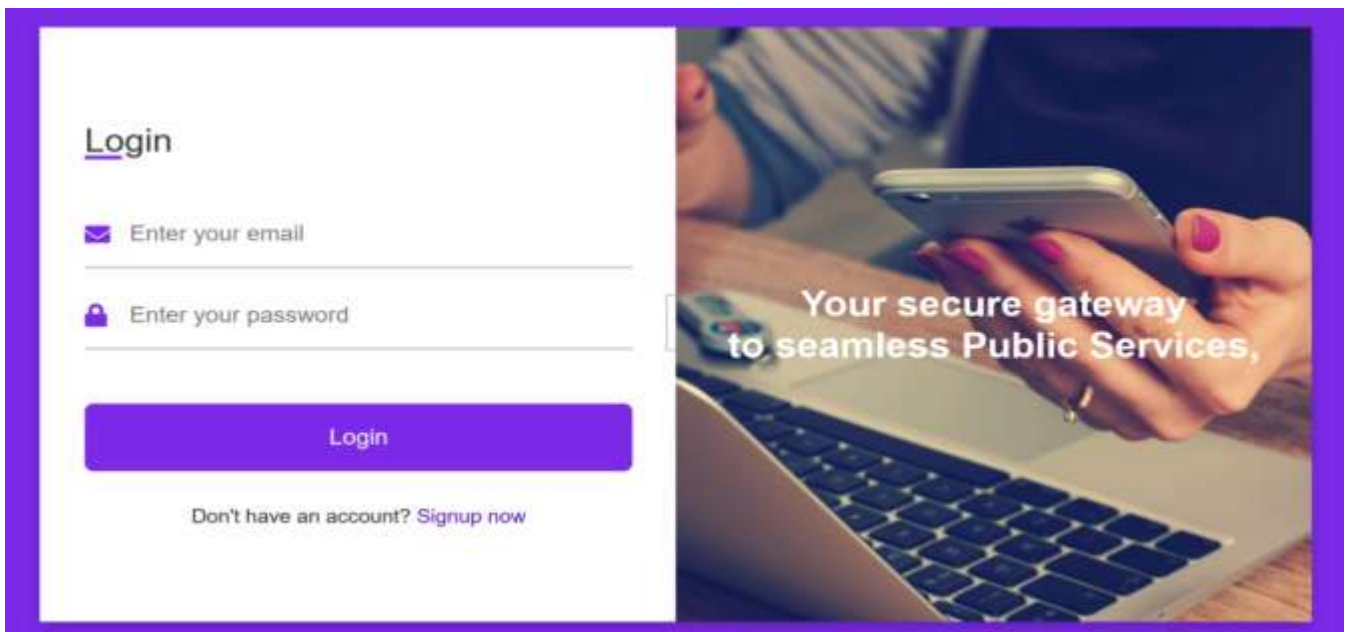
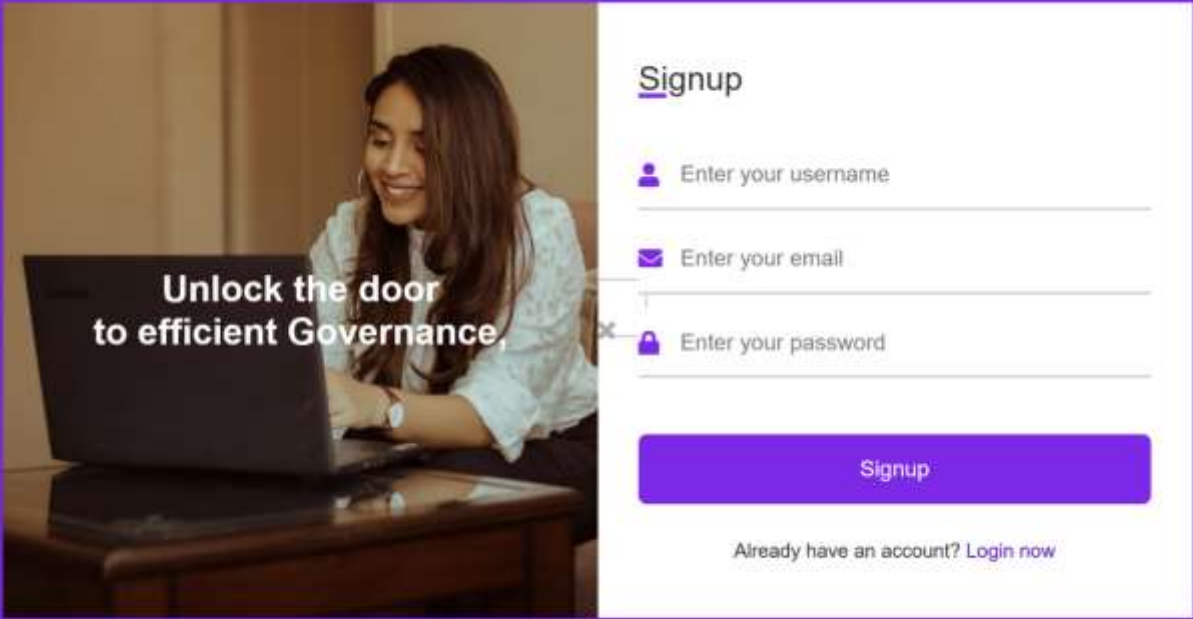


Figure 28: Login page

Appendix C: Signup Page



The image shows a web page layout for a signup form. On the left, a woman with long dark hair is sitting at a desk, smiling and looking at a laptop. The laptop screen displays the text "Unlock the door to efficient Governance,". On the right, there is a white rectangular box containing the signup form. The form has a title "Signup" at the top. Below the title are three input fields, each with a purple icon and a label: a person icon for "Enter your username", an envelope icon for "Enter your email", and a padlock icon for "Enter your password". Each input field has a horizontal line below it. Below these fields is a large purple button with the text "Signup" in white. At the bottom of the form box, there is a link that says "Already have an account? Login now".

Signup

Enter your username

Enter your email

Enter your password

Signup

Already have an account? [Login now](#)

Figure 29: Signup page



LOGOUT

Predict Demand for Huduma Service

Revolutionizing Public Service Delivery with Data
Analytics



Citizens queuing for Huduma Service

Fill the Form to Predict Demand

Quarter:

Select Quarter

Year:

Select Year

Service Name:

Select Service

Predict Demand

Figure 30: User interface to take user inputs to predict

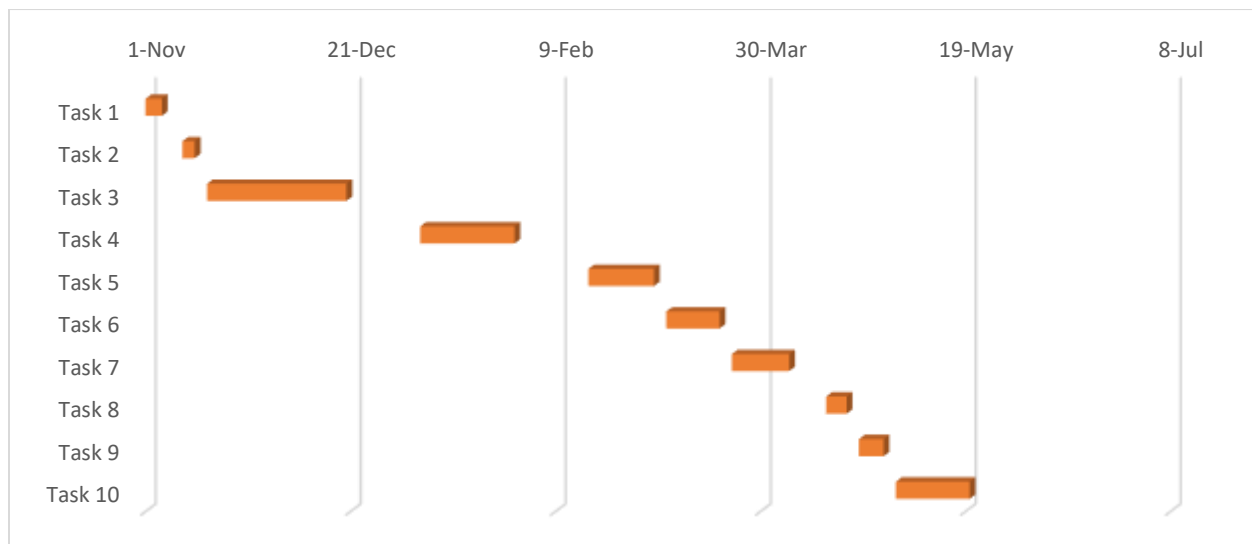
APPENDIX E: PROJECT PLAN

Table 3: Project plan

Task No.	Task Name	Planned Hours	Actual Hours	Planned Start Date	Actual Start Date	Planned End Date	Actual End Date	Deliverables
1	Define project scope and objectives	15	18	11/1/2023	11/4/2023	11/5/2023	11/5/2023	Project Plan and Project Charter
2	Project Planning	16	20	11/10/2023	11/10/2023	11/13/2023	11/13/2023	
3	Data Collection	10	14	11/16/2023	11/16/2023	12/15/2023	12/20/2023	Data Collection Report
4	Data Preprocessing	20	20	1/7/2024	1/20/2024	1/14/2024	1/30/2024	Preprocessed Data
5	Model Development	17	20	2/17/2024	2/20/2024	2/22/2024	3/4/2024	Candidate Model
6	Fine tuning Model	18	20	3/7/2024	3/8/2024	3/10/2024	3/12/2024	Refined Model
7	User Interface Development	17	20	3/15/2024	3/17/2024	3/18/2024	3/23/2024	System design document
8	Integrate Model with UI	16	20	3/25/2024	3/30/2024	4/12/2024	4/29/2024	Demand Prediction System
9	Documentation	18	20	5/2/2024	5/8/2024	5/13/2024	5/20/2024	Project Report

APPENDIX F: PROJECT GANTT CHART

Table 4: Project Gantt Chart



APPENDIX G: BUDGET ESTIMATES

Table 5: Budget Estimates

Hardware:	Costs (Kshs.)
Networking infrastructure	80,000
Computers (3 Pieces)	200,000
Total Hardware Costs:	315,000
Software:	Costs: (Kshs.)
Python	-
Data	20,000
IDE	-
Internet	60,000
Total Software Costs:	80,000
Personnel:	Costs: (Kshs.)
Data Scientists:	80,000
Machine learning Engineer:	80,000
Python Developer	50,000
System Administrator	55,000
Technical Writer:	20,000
Total Personnel	285,000
Total Budget	680,000

APPENDIX H: System Code

Model_training.py

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

import joblib

import pickle


def load_and_transform_data(csv_file_path, chunk_size=10000):

    transformed_chunks = []

    for chunk in pd.read_csv(csv_file_path, chunksize=chunk_size):

        demand_counts = chunk.groupby(['year', 'quarter',
'servicename']).size().reset_index(name='demand')

        transformed_chunks.append(demand_counts)

    data = pd.concat(transformed_chunks, ignore_index=True)

    return data


def clean_data(data):

    data = data[(data['year'] != 'quar') & (data['servicename'] != 'servicename')]

    data['year'] = pd.to_numeric(data['year'], errors='coerce')

    data = data.dropna(subset=['year'])

    data['year'] = data['year'].astype(int)
```

```

    return data

def encode_categorical_columns(data, categorical_columns):

    label_encoders = {}

    for column in categorical_columns:

        label_encoders[column] = LabelEncoder()

        data[column] = label_encoders[column].fit_transform(data[column])

    joblib.dump(label_encoders, 'label_encoders.pkl')

    return data, label_encoders

def feature_engineering(data):

    data['moving_avg'] = data.groupby('servicename')['demand'].transform(lambda x: x.rolling(4,
min_periods=1).mean())

    data['yoy_change'] = data.groupby(['servicename', 'quarter'])['demand'].transform(lambda x:
x.diff(periods=4))

    data['qoq_change'] = data.groupby(['servicename'])['demand'].transform(lambda x: x.diff())

    mean_yoy = data['yoy_change'].mean()

    mean_qoq = data['qoq_change'].mean()

    data['yoy_change'] = data['yoy_change'].fillna(mean_yoy)

    data['qoq_change'] = data['qoq_change'].fillna(mean_qoq)

    data['demand_lag1'] = data['demand'].shift(1)

    data['demand_lag2'] = data['demand'].shift(2)

    data.dropna(inplace=True)

    return data

def train_model(X_train, y_train):

```

```

rf_regressor = RandomForestRegressor(n_estimators=100, random_state=42)

rf_regressor.fit(X_train, y_train)

return rf_regressor

def evaluate_model(rf_regressor, X_test, y_test):

    y_pred = rf_regressor.predict(X_test)

    mse = mean_squared_error(y_test, y_pred)

    mae = mean_absolute_error(y_test, y_pred)

    r2 = r2_score(y_test, y_pred)

    return mae, mse, r2

def save_model(model, file_name='rf_regressor_model.pkl'):

    with open(file_name, 'wb') as model_file:

        pickle.dump(model, model_file)

def main():

    # Load and transform the data

    csv_file_path = r'data\transformed_data.csv'

    data = load_and_transform_data(csv_file_path)

    # Clean the data

    data = clean_data(data)

    # Encode categorical columns

    categorical_columns = ['quarter', 'servicename']

    data, label_encoders = encode_categorical_columns(data, categorical_columns)

    # Perform feature engineering

    data = feature_engineering(data)

```

```

# Separate features and target variable

X = data.drop('demand', axis=1)

y = data['demand']

# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train the Random Forest Regressor

rf_regressor = train_model(X_train, y_train)

# Evaluate the model

mae, mse, r2 = evaluate_model(rf_regressor, X_test, y_test)

print(f"Mean Absolute Error: {mae:.2f}")

print(f"Mean Squared Error: {mse:.2f}")

print(f"R-squared Score: {r2:.2f}")

# Save the trained model

save_model(rf_regressor)


if __name__ == "__main__":

    main()

```

managers.py

```

from django.contrib.auth.base_user import BaseUserManager

class UserManager(BaseUserManager):

    use_in_migrations = True

    def create_user(self, username, email, password=None, **extra_fields):

```

```
if not username:

    raise ValueError('The Username field must be set')

if not email:

    raise ValueError('The Email field must be set')

email = self.normalize_email(email)

user = self.model(username=username, email=email, **extra_fields)

user.set_password(password)

user.save(using=self._db)

return user

def create_superuser(self, username, email, password=None, **extra_fields):

    extra_fields.setdefault('is_staff', True)

    extra_fields.setdefault('is_superuser', True)

    return self.create_user(username, email, password, **extra_fields)
```